Plumbing and wet-room study · round 1 · inventory, overlay, constraints, scores, alternatives, three scenarios, the utility room · Taif / Palm House V8.1 · STUDY ONLY

Design review E-03, G-07 · open item 10 · 2 Nov 2026 · the latest geometry (the flipped stair, the family bath F-b, the south wing v2, the maid room and kitchen changes, the site plan's manholes and septic route) · **every size is assumed, to be designed by the plumbing engineer** · nothing in the design files changes until the owner decides

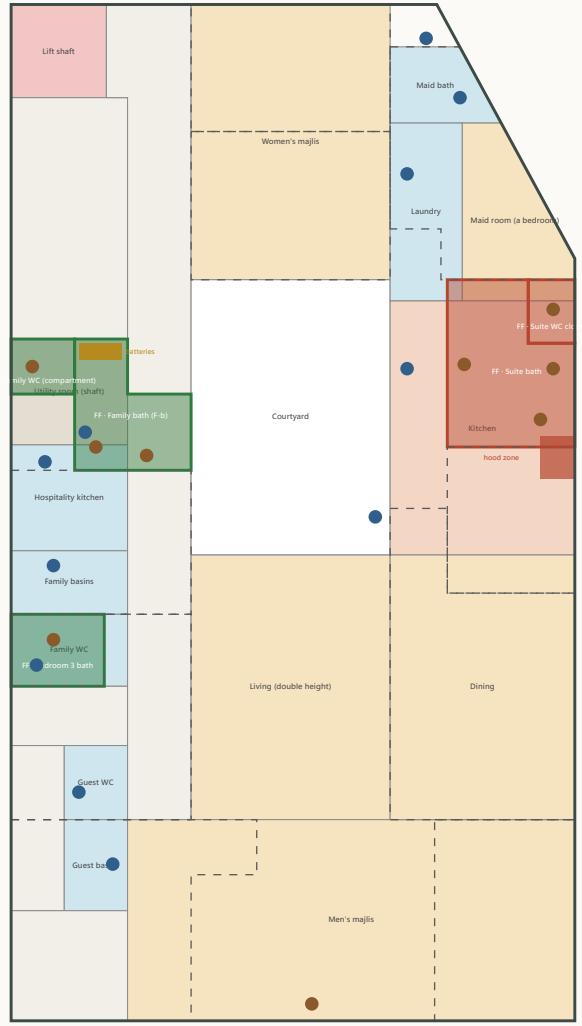
Principles applied in every scenario (owner's goals: maintenance without breaking floors or walls, no smells, low leak risk, quality, controlled cost)

1. **Two clusters only:** west round the shaft (the utility room, the hospitality kitchen, the family basins and WC, the guest WC and basins; above: the family bath F-b and its WC compartment, bedroom 3's bath) and east along the east wall (the kitchen, the laundry, the maid bath; above: the suite bath and its WC closet). **No third cluster** (bedroom 1 stays without a bath: decided).
2. **Ground floor:** the drains buried under the ground slab, a **cleanout at every change of direction**, an **inspection chamber outside each cluster** (the west manhole on the driveway, the east manhole in the yard), reachable from above; then the collector manhole and the existing septic (site plan).
3. **Stacks:** vertical stacks inside the **west shaft** (doors on each floor) or an **accessible duct** (scenario C: an internal east duct); acoustic (triple-layer) pipes on every stack; the east stack today runs outside the east wall behind a removable cladding.
4. **First-floor bathrooms:** a **sunken waterproofed slab** (0.10–0.15 m, the leak kept inside the bathroom) wherever the bay allows it (the ribbed bays; the suite bath has it, X10), OR suspended drains only above **service rooms and corridors** behind access panels with acoustic lagging; **never** above bedrooms, the majlis, the living or the kitchen's hood zone.
5. **No smell:** real **vent stacks to the roof** (the west shaft and the east stack; ≥ 3 m from the HRV intake and the malqaf), **deep-seal traps** (≥ 75 mm) on every fixture, **anti-odour floor drains** (membrane or ball type, with a primer) on the rarely used drains: the courtyard, the suite court, the laundry and the utility room, the breeze channel's sump.
6. **Water:** a **manifold system** (one continuous PEX pipe from the manifold to each fixture, no buried joints), a **manifold cabinet per floor near the shaft** with a valve per fixture (west: the utility room GF, the family bath's wall FF; east: the laundry GF, the suite WC closet's wall FF); **one water heater per cluster** close to its bathrooms (west: the utility room; east: the laundry, serving the kitchen, the maid bath and the suite bath above); a **booster pump** from the tank (site plan) with a main sediment filter in the utility room; an **RO unit** at the kitchen sink.
7. **Leak protection:** leak sensors under the heaters, the washer, the dishwasher and every basin cabinet, wired to an **automatic main shut-off valve** after the filter; drip trays under the heaters and the washer, drained to a floor drain; the first-floor bathrooms' sunken slabs are the second line.

1 · Inventory: every wet point (both floors and the roof)

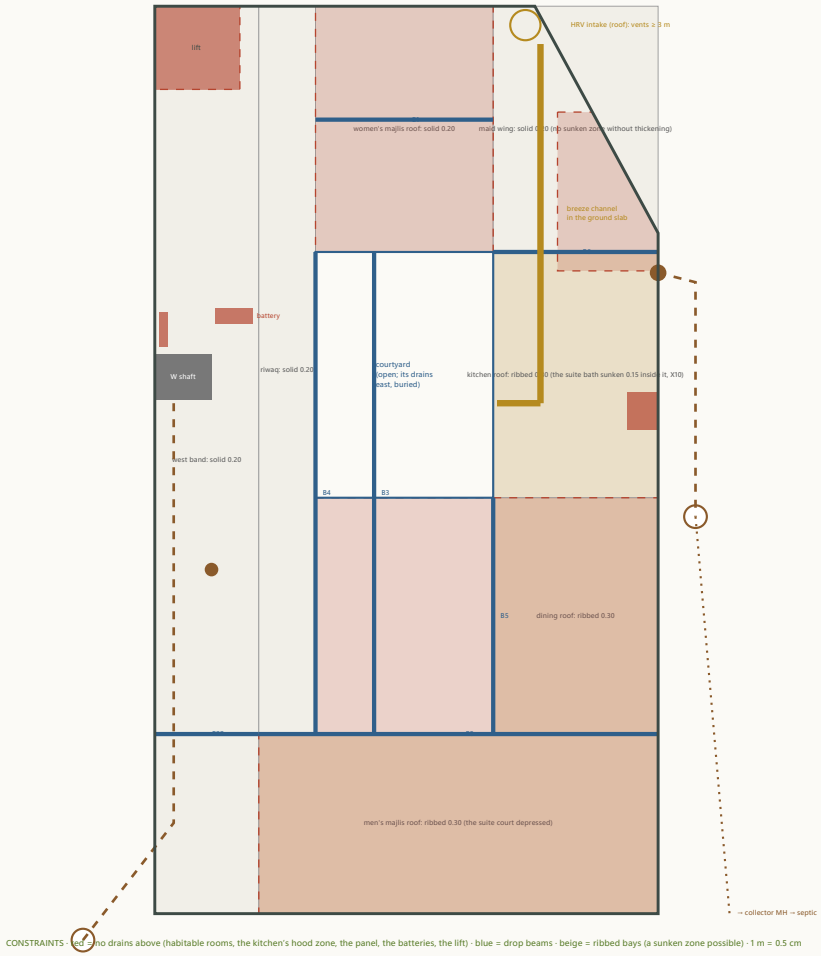
Fl.	Room	Fixtures	Drains to	Branch m	Use
GF	Hospitality kitchen	sink	W-RUN	0.3	daily
GF	Family basins	2 basins (wudu)	W-RUN	0.5	daily
GF	Family WC	WC + basin	W-RUN	0.1	daily
GF	Guest WC	WC + basin	W-RUN	1.1	guests
GF	Guest basins	trough, 3 taps	W-RUN	3.2	guests
GF	Utility room	heater drip tray · condensate	W-SHAFT	0.3	rare
GF	Kitchen	sink + dishwasher	E-MH	9	daily
GF	Laundry	washer · dryer · utility sink · floor drain	E-MH	13.5	daily / rare (floor drain)
GF	Maid bath	WC · basin · shower	E-MH	13.2	daily
GF	Courtyard	step drain · overflow · basin + planter	RAIN	6.2	rain / washing
GF	Air tower	breeze channel sump · HRV condensate	E-MH	14.5	rare
FF	Family bath (F-b)	shower	W-SHAFT	0.6	daily
FF	Family bath (F-b)	double basin	W-SHAFT	1.7	daily
FF	Family WC (compartment)	WC	W-SHAFT	1.2	daily
FF	Bedroom 3 bath	shower · WC · basin	FWC	0.5	daily
FF	Suite WC closet	WC	E-STACK	0.5	daily
FF	Suite bath	shower (east wall)	E-STACK	1.6	daily
FF	Suite bath	tub (by the east window)	E-STACK	3.1	weekly
FF	Suite bath	double vanity (west wall)	E-STACK	3	daily
FF	Suite court	3 linear drains · box gutter · overflow	RAIN	3.2	rain
roof	Roof	11 outlets → internal downpipes	RAIN	—	rain
roof	Roof (plant)	AC condensate (Z2b) · HRV condensate (base)	E-STACK	—	daily (summer)

Branch = the horizontal drain from the fixture to its collector (the buried run, a stack or the manhole), measured on the routes drawn; the roof outlets (11) and the AC / HRV condensate are listed, not scored. Total of the scored branches: 77.3 m.



OVERLAY · the first-floor wet rooms (green = over a wet / service room, red = over the kitchen) on the ground floor · dots: wet points (blue GF, brown FF) · 1 m = 0.56 cm

2 · Constraints map · what lies under each first-floor wet room · the stacks and the routes to the septic



CONSTRAINTS · red = no drains above (habitable rooms, the kitchen's hood zone, the panel, the batteries, the lift) · blue = drop beams · beige = ribbed bays (a sunken zone possible) · 1 m = 0.5 cm

No-drain zones above

- **Habitable rooms (no drains in their ceilings):** the women's majlis, the men's majlis, the living and the dining, the maid room (a bedroom); on the first floor every bedroom, the kids' room, the family living, the office and the gym carry only the roof (no first-floor fixtures above them). **The kitchen:** its **hood zone** on the east wall (y 10.2–11.2) takes no drain above it; the rest of the kitchen's ceiling carries the suite bath only inside a sunken tanked slab (no suspended pipes in the kitchen's ceiling). **The electrical panel and the battery space** (the utility room's north part) and the **lift shaft**: nothing above or beside.

Structure

- **Drop beams** (blue): B1 (y 3.0), B4 (x 4.25), B5 (x 8.95 south of y 13), B6 (y 6.5), B8 / B22 (y 19.25), the edge beam B3: a drain never crosses a drop beam; branches run parallel to them or through sleeves cast in the hidden beams (the engineer).
- **Slabs:** the west band, the riwaq, the maid wing and the women's majlis roof are **solid 0.20**: a sunken bathroom zone there means a thicker bay (0.35 with the top depressed 0.15) or a raised floor; the kitchen, dining and men's majlis roofs are **ribbed 0.30**: a 0.10–0.15 depressed zone is possible (the suite bath has one). The family bath F-b over the utility room / hospitality kitchen is on the solid west band: its sunken zone needs the thicker bay (X10-type detail) or its drains stay suspended above the two **service** rooms behind panels (allowed by principle 4).
- **The malqaf breeze channel** (amber) in the ground slab from the tower south on x 10.2 to y 10.5 then west under the kitchen's sink counter: the kitchen's drain crosses it (the sink's branch passes **under** the channel's base slab, or the channel bridges it: the engineers); the **foundations** (pads ≈ 1.5 × 1.5 under the columns, strip footings under the RC walls): the buried drains pass between them; the **courtyard**: its drains run east under the kitchen floor to the east manhole's rain branch.

The two stacks and the routes to the septic

- **West:** the shaft (0–1.5 × 9.2–10.4) carries the west stack and vent from the family bath / WC compartment down to the buried west run, which collects the hospitality kitchen, the family basins, the family WC (its own small stack takes bedroom 3's bath from above) and the guest WC and basins, and leaves under the west wall to the **west manhole** on the driveway. **East:** the stack outside the east wall beside the suite WC closet takes the suite bath, drops to the yard and runs south to the **east manhole** (foul only), which also takes the kitchen, the laundry and the maid bath (buried, east); the courtyard's rain goes to its own chamber (round 2, A2). Both manholes run to the **collector manhole** at the garden's south-east and the **existing septic** (site plan; the capacity, the condition and the well distance flagged).

Overlay verdicts

First-floor wet room	Below it	Verdict
Family bath (F-b)	Utility room (shaft), Hospitality kitchen, Lobby + riwaq	good: over wet / service
Family WC (compartment)	Utility room (shaft)	good: over wet / service
Bedroom 3 bath	Family WC	good: over wet / service
Suite bath	Maid room (a bedroom), Laundry, Kitchen	problem: over the kitchen
Suite WC closet	Maid room (a bedroom), Kitchen	problem: over the kitchen

3 · Score of every wet room · 4 · Alternatives for the rooms that score poorly

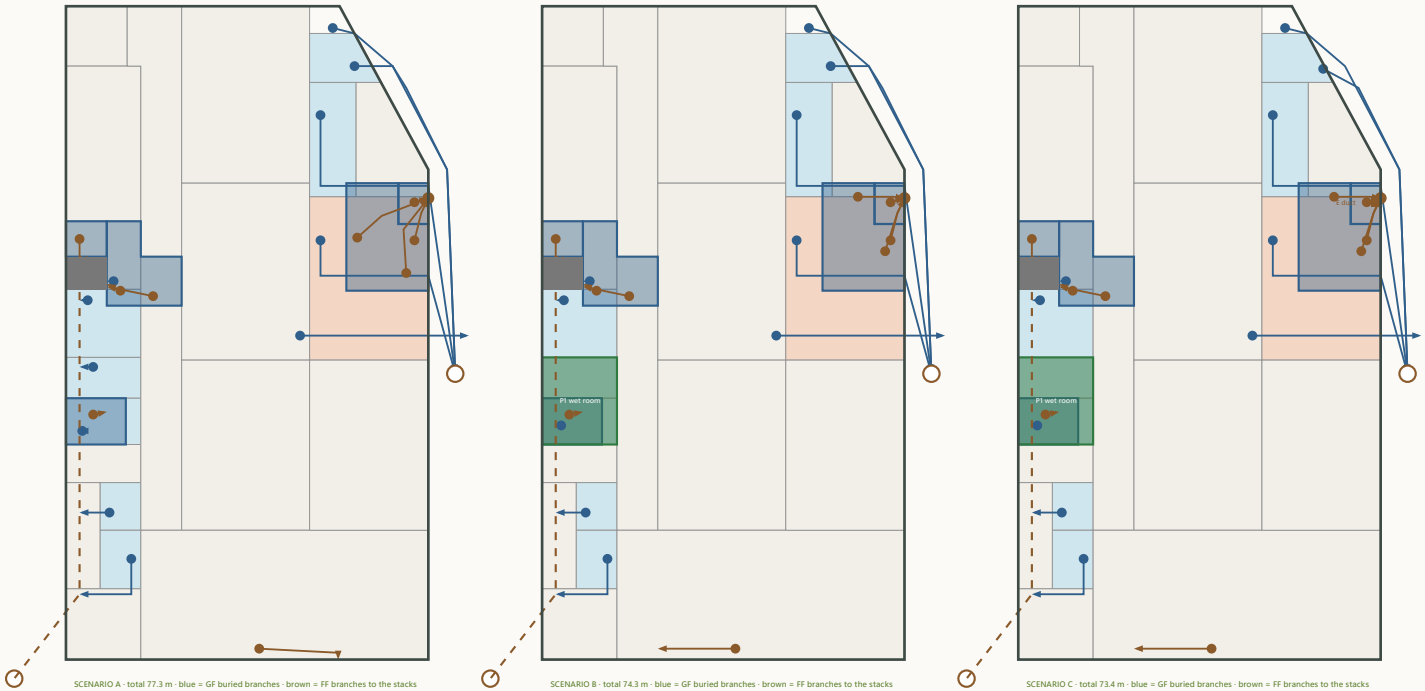
Fl.	Room	Fixture	m	bends	over	access	smell	score / 10
GF	Hospitality kitchen	sink	0.3	0	buried	cleanout	trap + vent	9.9
GF	Family basins	2 basins (wudu)	0.5	0	buried	cleanout	trap + vent	9.8
GF	Family WC	WC + basin	0.1	0	buried	cleanout	trap + vent	10
GF	Guest WC	WC + basin	1.1	0	buried	cleanout	trap + vent	9.5
GF	Guest basins	trough, 3 taps	3.2	1	buried	cleanout	trap + vent	7.7
GF	Utility room	heater drip tray · condensate	0.3	0	buried	shaft door	anti-odour drain	8.9
GF	Kitchen	sink + dishwasher	9	2	buried	cleanout	trap + vent	5.6
GF	Laundry	washer · dryer · utility sink · floor drain	13.5	2	buried	cleanout	anti-odour drain	4.6
GF	Maid bath	WC · basin · shower	13.2	3	buried	cleanout	trap + vent	5
GF	Courtyard	step drain · overflow · basin + planter	6.2	1	buried	cleanout	anti-odour drain	5.3
GF	Air tower	breeze channel sump · HRV condensate	14.5	3	buried	cleanout	anti-odour drain	4
FF	Family bath (F-b)	shower	0.6	0	wet / service	shaft door	trap + vent	9.7
FF	Family bath (F-b)	double basin	1.7	0	wet / service	shaft door	trap + vent	9.1
FF	Family WC (compartment)	WC	1.2	1	wet / service	shaft door	trap + vent	8.7
FF	Bedroom 3 bath	shower · WC · basin	0.5	0	wet / service	stack panel	trap + vent	9.7
FF	Suite WC closet	WC	0.5	0	kitchen	outside stack	trap + vent	6.7
FF	Suite bath	shower (east wall)	1.6	1	kitchen	outside stack	trap + vent	5.5
FF	Suite bath	tub (by the east window)	3.1	1	kitchen	outside stack	trap + vent	4.8
FF	Suite bath	double vanity (west wall)	3	2	kitchen	outside stack	trap + vent	4.1
FF	Suite court	3 linear drains · box gutter · overflow	3.2	1	majlis roof (rain)	outside stack	anti-odour drain	4.7

Score = branch length (3) + bends (2) + stacked over a wet / service room (2) + maintenance access (1.5) + smell risk (1.5); structural impact and cost are in the text; the weights are this study's, not a code. Mean today: 7.2.

The rooms that score poorly, and their alternatives

- **The suite bath and its WC closet (over the kitchen):** the only first-floor wet room over a problem room, and the house's longest first-floor branch (the vanity on the west wall, 3 m). **Alternatives:** (a) **keep it** with its sunken tanked slab 0.15 (exists, X10), every branch inside the sunken zone, the vanity branch along the north wall; nothing in the kitchen's ceiling; (b) **move the vanity to the east side** beside the stack (2.1 m) and the tub to the north wall (1.7 m): the branches shorten, the room keeps its function (a double vanity still faces the east light from the window); (c) **swap the bath and the dressing** (the bath over the kitchen's south part / the dining): still over the kitchen and farther from the stack: worse; (d) **move the bath over the laundry / maid wing** ($y \le 7.0$): the suite lobby and bedroom 1 are there: not possible without re-planning the suite. **Chosen for B and C: (b).**
- **The suite court's gutter outlets** (rain, in the men's majlis's south wall): (a) keep, with acoustic lagging; (b) run the two outlets to the court's ends and down inside the **gym's and the office's** walls to the garden: no pipe in a majlis wall. **Chosen for B: (b).**
- **The courtyard's drain, overflow and basin** (6.2 m east under the kitchen floor to its own rain chamber, never the foul one): buried, fine; anti-odour drain and the basin's own trap; a cleanout at the kitchen's west wall. Keep.
- **The maid bath** (13.2 m round the cut corner): (a) keep; (b) **align its fixtures on the cut wall** with one short branch out through the cut wall to the strip, and the breeze channel's sump and the HRV condensate into the same branch. **Chosen for C.**
- **The ground-floor shower** (G-07, deferred to this study): **P1** (the family WC + basins as one wet room on the west run, 0.5–1.5 m branches, a level floor drain in the ground slab: easy) scores as well as any fixture here; **P2** (the hospitality kitchen, 1 m from the shaft) is the plumbing optimum but re-plans three rooms; **P1-lite** (the family WC room only) is P1 without the basins. **Chosen for B and C: P1**, for its function as much as its plumbing.
- **The laundry:** fine (buried east); C adds the east heater and manifold cabinet beside the washer and moves the utility sink next to it (one trap group).

4 · Three whole-house scenarios · A minimum · B moderate · C ideal · the comparison and the recommendation



Criterion	A · minimum	B · moderate	C · ideal
What changes	nothing moves; cleanouts, two vent stacks, anti-odour drains, acoustic stacks, the manifold system, two heaters, leak protection	A + the GF shower P1 (the family WC + basins → one wet room), the suite vanity beside the stack and the tub on the north wall, the court's outlets down the gym / office walls	B + an internal accessible east duct (0.35 × 0.35, the kitchen's NE corner under the suite WC closet) with doors on both floors; the maid bath on the cut wall; the east heater + manifold in the laundry
Total horizontal drain (scored branches)	77.3 m	74.3 m	73.4 m
Longest first-floor foul branch	3.1 m (the suite vanity)	2.1 m (the suite vanity)	2.1 m
Wet rooms over a problem room	3 (the suite bath over the kitchen: sunken tanked slab)	3 (the same)	3 (the same; its branches now reach an internal duct)
Maintenance access	the west shaft (doors), the east stack outside behind cladding, cleanouts	as A; the GF wet room's floor drain from above	every stack behind a door on each floor; no outside pipe
Smell risk	low: vents, deep traps, anti-odour drains	low	lowest: the rare drains grouped on the east duct with a primer
Structural changes	none (the suite bath's sunken zone exists)	B14 under the removed wall of the family WC (a drop beam, the shower study)	B + a 0.35 × 0.35 opening through the kitchen roof slab beside C33 and the east wall (the engineer)
Rough relative cost	1.0 (the base: pipes, stacks, manifolds, 2 heaters, sensors)	≈ 1.15 (one wet room re-fitted, a beam, a window)	≈ 1.3 (the duct, the maid bath re-fit, the laundry re-fit)
Function impact	none	the wudu basins behind a sliding door; the suite vanity on the east side (lit by the window)	B + the maid bath re-planned (same size), the laundry loses 0.6 m of counter
Fixtures scoring below 5 / mean	5 of 20 / 7.2	2 of 19 / 7.2	2 of 19 / 7.2

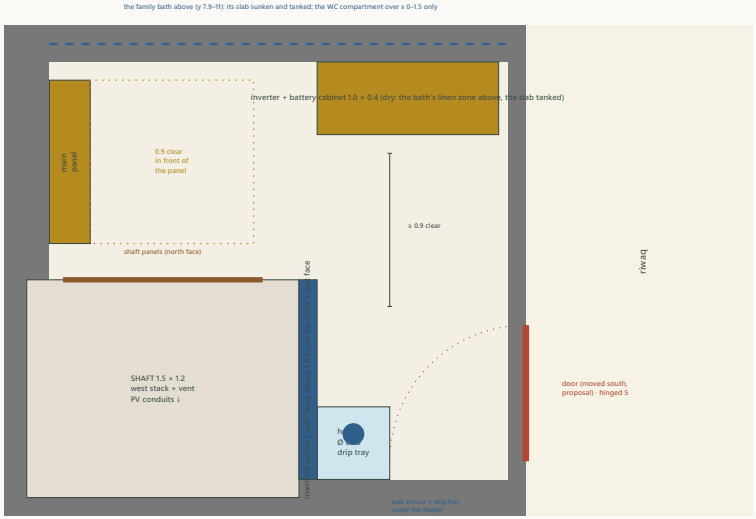
Recommendation: B, plus C's internal east duct if the engineer allows the slab opening

- The house is already well clustered: every first-floor wet room but one stands over a wet or service room, and the one exception (the suite bath over the kitchen) is mitigated by its sunken tanked slab and cannot be moved without re-planning the suite. So the gains lie in the **routes, the access and the venting** (A) and in a few short moves (B): the GF shower as P1, the suite vanity beside the stack, the court's outlets out of the majlis wall. B costs little and removes the longest branch.
- C's one real gain is the **internal east duct**: it brings the east stack indoors behind a door on each floor (maintenance without breaking walls, no pipe on the facade) at the price of a slab opening in the kitchen's corner; the maid bath and laundry re-fits are small. **Take the duct if the engineer accepts the opening; leave the rest of C.**
- **In every scenario:** the sunken slabs are the second line against leaks; the manifold system and the sensors with the automatic valve are the first; the two vent stacks run to the roof ≥ 3 m from the HRV intake (12.5 / 7.4 m).

Decisions needed from the owner

1. **Scenario:** A, B (recommended) or C; and, with B, **the internal east duct** from C (yes / no, subject to the engineer).
2. **The GF shower:** P1 confirmed as the wet room of the west cluster (or P1-lite / P2)?
3. **The suite bath:** the vanity moved to the east side beside the stack (B) — accepted?
4. **The suite court's outlets** down the gym / office walls instead of the majlis wall (B)?
5. **Heaters:** two electric cylinders (west ≈ 100–120 L in the utility room, east ≈ 150 L in the laundry), or a central heat-pump unit on the roof with a ring main (less space inside, more pipe)? (to be sized by the engineer)
6. **The utility room's door** moved south (y 9.45–10.2, 0.75 wide) to free the north wall for the panel and the batteries (page 5)?

The utility room (west cluster): the shaft, the panel, the battery space, the heater, the manifold, the access clearances · the water kept away from the batteries



UTILITY ROOM · 2.525 × 2.3 clear · 1 m = 2.4 cm · north up

Layout (proposal, every size assumed)

- **The room:** x 0–2.75, y 7.9–10.4 on axes, **2.525 × 2.3 m clear**; the **shaft** 1.5 × 1.2 in its south-west corner (the west stack and vent, the PV conduits from the roof, the cold and hot risers to the first floor), its **access panels on the north face** (removable, 1.1 m wide) so the shaft is serviced from the room's north part; the **door** moved from y 8.2–9.0 to **y 9.45–10.2** (0.75 wide, hinged on the south jamb, opening west: its swing stays south of the battery wall), a proposal for `design_gf.typ`.
- **Dry side (north and west):** the **main electrical panel** on the west wall (x 0.125–0.35, y 8.1–9.0) with 0.9 m clear in front; the **inverter and the battery cabinet** 1.0 × 0.4 on the **north wall's east part** (x 1.6–2.6): nothing wet above or beside it — above it lies the family bath's **dry linen zone** (x 1.5–2.75, y 7.9–9.2), the WC compartment sits over x 0–1.5 only, and the bath's slab is sunken and tanked; the heater is 1.5 m away, the manifold 0.8 m, with a drip tray under each. The alarm panel and the recorder (site plan) share the dry side.
- **Wet side (south):** the **water heater** (an electric cylinder Ø 0.40, ≈ 100–120 L, floor-standing on a drip tray with a leak sensor) at x 1.6–2.0 beside the shaft's east face; the **manifold cabinet** wall-hung above 1.8 m on the shaft's east face (x 1.5–1.6, y 9.2–10.3, above the heater's top) with the valves, the main sediment filter, the booster pump's connection and the automatic shut-off valve; the floor drain by the heater into the west stack.
- **Clearances:** 0.9 m in front of the panel **OK**, the battery's front free to the door swing **OK**, the heater reachable from the room's middle, the shaft panels reachable from the north part; the free floor ≈ 4.3 m².

Flags

- **The room is small:** with the shaft it leaves 4.3 m² of free floor; a 200 L cylinder or a heat-pump unit does not fit (a roof heat pump would); the batteries and the water share one room by the owner's decision, with the layout above as the condition. **The electrical engineer confirms** the battery cabinet's space, its ventilation (a lithium cabinet, ventilated to the riwaq or the shaft), its fire safety (a steel cabinet, ≥ 0.5 m from combustibles, the smoke alarm by the stair, ideally a 60-minute separation to the stair store) and the panel's position.
- **If the engineer wants more:** the store under the stair's half landing (north of this wall, 2.525 × 1.25, 1.65 m high, the stair above it, dry) can take the battery cabinet instead, with a fire-rated door; the owner's decision today is the utility room.

Round 1 checks passed (9)

Round 1 checks: every first-floor wet room except the suite bath stands over a wet or service room (the overlay) · the suite bath and its WC closet stand over the kitchen (reported as the one unavoidable problem, mitigated by the sunken tanked slab) · no first-floor drain crosses the kitchen's hood zone (the tub's branch kept north of y 10.2 at the east wall) · the two vent stacks ≥ 3 m from the HRV intake (west 12.5 m, east 7.4 m) · every first-floor foul branch ≤ 4 m horizontal (max 3.1 m, the suite vanity today) · scenario B shortens the longest first-floor foul branch to 2.1 m and the total to 74.3 m (A: 77.3), and leaves fewer fixtures below 5 (2 vs 5) · utility room: the battery cabinet not under the WC compartment (x ≥ 1.5), ≥ 1 m from the heater (1.5) and ≥ 0.75 m from the manifold (0.8) · utility room: the door's swing clear of the heater, the battery and the manifold; 0.9 m in front of the panel; the shaft panels on its north face reachable · utility room: free floor ≈ 4.3 m² (small: reported). Every size is assumed, to be designed by the plumbing engineer; nothing here changes the room CSVs or the design files. Round 2 (3 Nov 2026) follows on pages 6–11.

Round 2 (3 Nov 2026) · the owner's provisional baseline · corrections: rain and sewage fully separate; condensate and the tower sump

The baseline for round 2 (owner, provisional)

- **Scenario B** is the direction: the ground-floor shower **P1** (the family WC + basins as one wet room) and the **suite court's outlets down the gym's and the office's walls**; the cleanouts, the two vent stacks, the anti-odour drains, the manifold system, two heaters, the leak protection as round 1.
- **The suite bathtub STAYS** on the south wall by the east window (round 1's "tub to the north wall" is dropped). The vanity's move "beside the stack" of round 1 clashed with the shower once the tub stays: **the vanity stays on the west wall**; how its 3 m drain is handled is the subject of A1 (page 7–9).
- Nothing is applied: the room CSVs and the design files are unchanged; every size assumed, to be designed by the plumbing engineer.

A2 · Rainwater and sewage fully separate (corrected)

- **Corrected:** round 1 and the site plan let the **courtyard's drain and overflow** reach the east manhole's "rain branch" beside the foul branches, and the site plan's manholes carried rain and foul to one collector. Now: the east yard chamber is **foul only**; the courtyard's outlet (x 13.311, y 12.1) runs to its **own rain chamber** in the strip (1.5 m) and the east soakaway; the **suite court's** three linear drains and box gutter leave down the **gym's and the office's walls** to the garden's south soakaway; the **roof's 11 outlets** discharge at the plinth to the driveway's channel and the two soakaways; the **site** surface water falls to the soakaways and the channel. **Nothing of the rain enters the septic**, and no foul pipe is tied to a rain pipe anywhere (the round-1 inventory: every "rain" point to RAIN, checked).
- **The final discharge** (the engineer, with the soil report): soakaways ≥ 5 m from the house's foundations and ≥ 10 m from the well (assumed rules), sized for Taif's storms; each with a **safe surface overflow** towards the lane side / the driveway's channel to the street, never towards the house; the garden's soakaway doubles as the palms' watering sump if the owner wishes.
- **The site plan is corrected the same way** (`studies/site_plan.typ` page 4: the rain chamber, the foul chambers, the septic now north-west near C1 and the inverts computed).

A2 · Condensate and the tower sump: the connection decided for each device

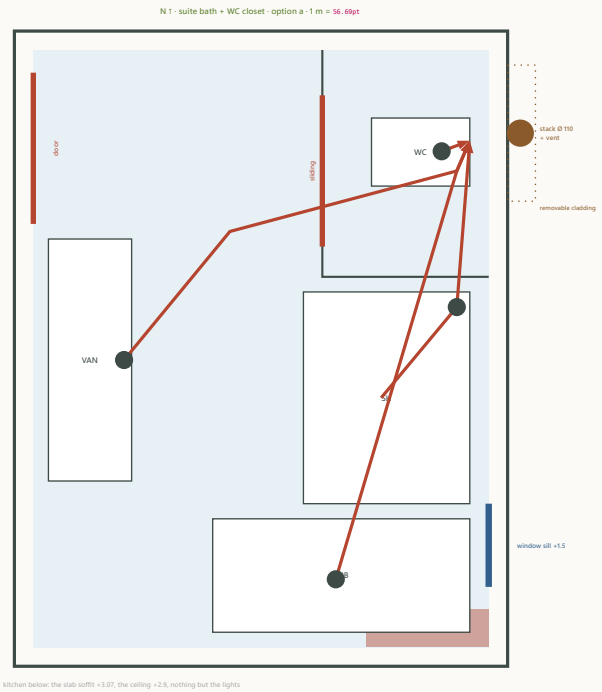
Device	Connection (never direct into a drain or a vent)	The trap kept wet	Cleaning
AC indoor units (fan-coils, FF ceilings / the roof zone Z2b)	Ø 25 PVC condensate pipe at 1 %, to a **tundish with an air gap** over a trapped waste in the nearest bathroom (the suite bath's service wall; the family bath's shaft) or to the roof's rain outlet (never a foul pipe directly)	the receiving trap is a daily-used fixture's (the vanity / the channel): never dries	the pan and the pipe flushed each season; the tundish is visible
the HRV unit (roof plant, its base)	Ø 25 to a **tundish with an air gap** over the roof's rain outlet (rain system: a condensate is clean water), with a dry-season drip tray	no trap needed on the rain side; the outlet's leaf guard kept clear	the HRV's drain pan cleaned at the filter change
the malqaf / tower sump (the breeze channel's base, GF)	a **trapped gully with an anti-odour membrane and a sealed access cover** at the channel's lowest point, drained by a **trapped air-break** (the gully's water seal + a 50 mm membrane valve) into the east foul branch (the maid bath's); it takes the channel's washing water and any wind-blown rain	the membrane valve closes when the seal dries; a **trap primer** from the nearest cold tap (the laundry) or a monthly bucket of water (the maintenance list)	through the tower's base door: the cover lifted, the gully basket and the membrane washed
the heaters' drip trays (utility room, laundry) and the washer's tray	the tray's Ø 32 to a floor drain with an **anti-odour membrane** (the west shaft's / the laundry's), an air gap at the tray	the floor drain primed by the tray's rare water: a membrane valve, not a water seal alone	the tray and the drain visible, cleaned with the room
the kitchen's RO unit (its reject water) and the dishwasher	to the sink's trap through an air-gap fitting (the RO) and a standpipe or the sink trap's spigot above its seal (the dishwasher): never a direct joint to the drain	daily use keeps the sink's trap wet	the air-gap fitting on the counter, the standpipe behind the dishwasher

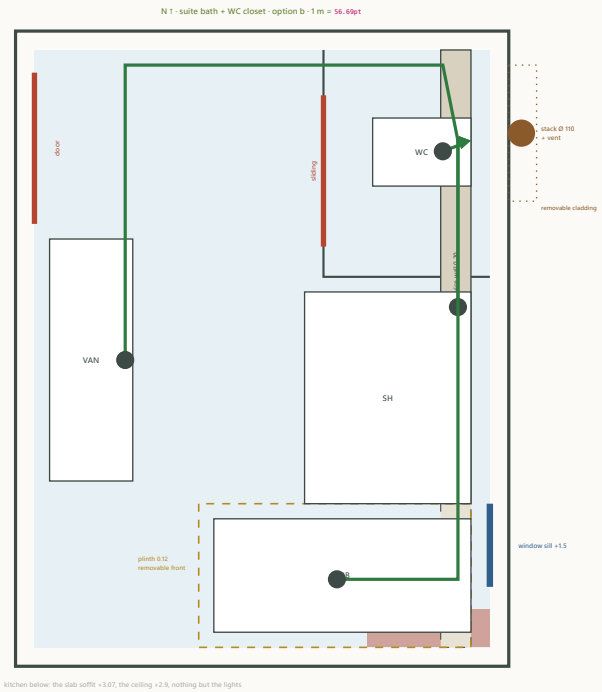
The detail (A2): a **tundish** is an open funnel with a visible air gap of ≥ 2 × the pipe's bore (≥ 50 mm) between the condensate pipe's end and the funnel's rim, draining to a trapped fixture: a blocked drain shows as a drip at the tundish, and no sewer gas can climb the condensate pipe. A **trapped air-break** is a gully whose water seal (50 mm) separates the device from the drain, with a **membrane (ball or flap) valve** that closes when the seal dries, and a **trap primer** (a small bleed from a cold tap through a vacuum breaker) or a listed monthly fill. Every size assumed, the plumbing engineer designs.

A2 · The site drainage corrected (summary of the site plan, page 4)

- **The septic** is **outside the farm wall near corner C1** (north-west, the north street), approximate; the foul drains of both clusters run **north and west**: the east main from the yard's foul chamber round the cut wall, along the north strip's north edge to the **collector chamber inside the gate**; the west branch from the driveway's west chamber (the west run now falls **north** inside the house); then under the wall west of the gate's post to the septic.
- **Falls (assumed):** the main at 1 %, the branches 1.5 %, 0.03 m through each chamber → the septic's inlet at −1.19 below the house floor, ≈ 0.79 m below the ground at the gate: **the septic's inlet level must be surveyed**; if it is higher, a packaged pump station at the collector chamber (flagged).
- **The tanks:** the existing one east of peg R, the reserved future tank beside it, ≥ 5 m from the house (20 m); the water pipe ≥ 1.5 m from the foul drains on the driveway; the septic ≥ 15 m from the tanks and ≥ 30 m from the well (rules assumed, to confirm by the engineer).

A1 · The suite bathroom over the kitchen · enlarged plans: (a) every drain in the sunken tanked slab (current) · (b) the east service wall + the tub’s plinth





The room and the stack

- The bath x 10.3–13.311, y 6.5–10.45 (3.01 × 3.95 on axes) with the WC closet in its north-east part (12.211–13.311 × 6.5–8.0, sliding door); the **stack outside the east wall** at y 7.05 behind a removable cladding panel, its branch fittings at the first-floor level, its vent to the roof; the **east window** y 9.5–10.05 (sill +1.5); the **hood zone** of the kitchen below starts at y 10.2 on the east wall: the depressed slab reaches y 10.45 but **no drain runs south of y 10.15** (the tub’s outlet at y 10 checked).
- **Fixtures** (`studies/ff_suite.typ`): the WC in the closet on the east wall (its outlet 0.2 m from the wall), the walk-in shower on the east wall y 8.1–9.5, the freestanding tub on the south wall by the window (its waste at x 12.3), the double vanity on the west wall (its trap 2.9 m from the stack by the floor route).

(a) all in the sunken tanked slab (current, red)

- The ribbed slab’s top is depressed 0.15 over the whole bath (+3.37); the build-up from it: the tanking membrane and its protection screed (+3.39), the **pipe zone 0.14 m**, a lightweight fill, the 50 mm screed, the second membrane under the tiles, the tiles (+3.6). The four drains lie in the fill at 2 % to the east wall and pass through it into the stack’s branches: all four fit (page 8), the WC’s Ø 110 with 0.02 m and the tub’s flat trap on its diagonal with 0.02 m to spare at the stack: tight.
- **Buried:** the WC’s bend, the shower’s gully and its outlet, the tub’s trap and its 3.1 m outlet, the vanity’s 3 m outlet and every joint of them. Reached only by lifting tiles, screed and the second membrane; the tanking under them stays.

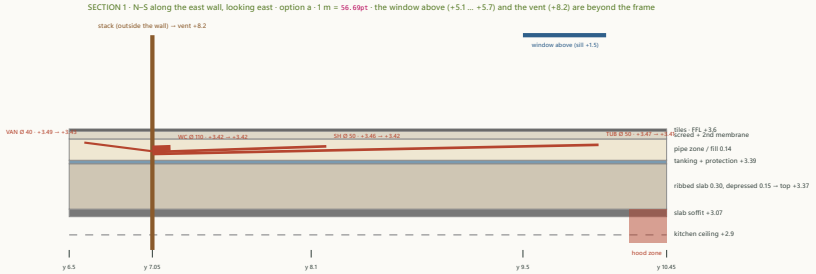
(b) the east service wall + the tub’s plinth (green)

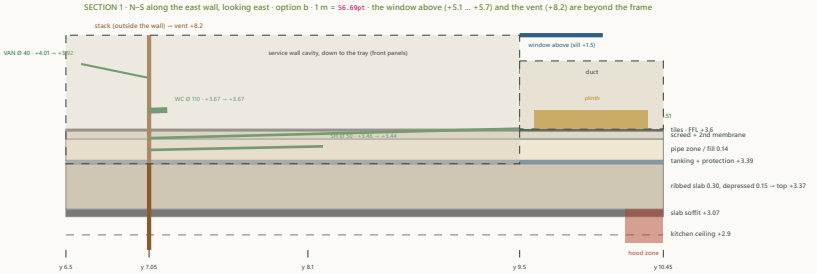
- A **0.20 m service wall** inside the east wall from the closet to the shower’s end (y 6.5–9.5, full height, its cavity down to the depressed slab), continued as a **0.45 m high duct** under the window to the south wall; the **WC wall-hung** on the wall’s frame with its outlet **horizontal at +0.18 straight through the wall into the stack** (0 m of floor pipe); the **shower channel along the service wall with its trap body inside the wall cavity**; the **tub on a 0.12 m plinth with a removable front** (its trap in the plinth’s void above the floor; the pipe in the plinth, the duct and the wall to the stack, 3.79 m at 2 %, arriving at +3.51, at about the floor level inside the wall cavity, 0.14 m above the tray); the **vanity’s Ø 40 in a chase in the north wall at +0.45** (one jointless pipe from its bottle trap) into the service wall. **What still lies in the floor:** the shower channel’s body only (set in the screed at the wall, its trap in the wall). Note: the door on the west wall (y 6.65–7.65) is not touched; the chase runs behind the plaster above the skirting.

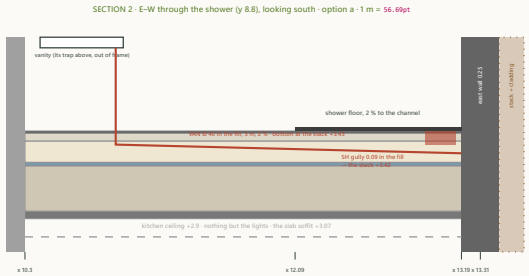
(c) a raised floor with removable panels

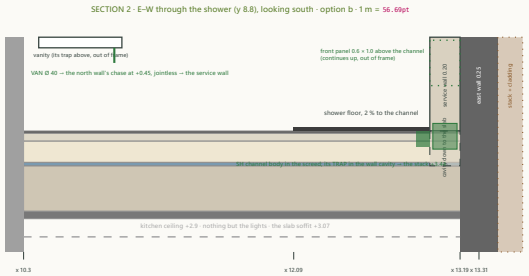
- A floor on pedestals over the tanked slab with lift-out tiles is practical only where no water falls on it: it cannot be sealed and removable at once in the shower. It works for the **tub’s zone** (= the plinth of (b)) and for a **dry strip along the west wall** under the vanity (0.6 wide, lift-out stone slabs over the vanity’s pipe): used in (b) as the plinth; the west strip is an option if the owner prefers the vanity’s pipe in the floor to a wall chase.

A1 · Dimensioned sections · the levels available in the floor build-up and the sunken slab against every drain · the kitchen ceiling below









Drain	Trap	OD	run m	bends	(a) crown at the fixture	(a) bottom at the stack	(a) fits	(b) bottom at the stack
WC	integral, 50 mm seal	Ø 110	0.2 / 0.2	0 / 0	+3.53	+3.42	yes, 0.02 spare	+3.67 (above +3.37)
SH	channel trap, 50 mm seal (+ membrane valve)	Ø 50	1.88 / 1.19	1 / 1	+3.51	+3.42	yes, 0.03 spare	+3.44 (above +3.37)
TUB	tub trap, 50 mm seal (+ AAV in option b)	Ø 50	3.03 / 3.79	1 / 2	+3.52	+3.41	yes, 0.02 spare	+3.51 (above +3.37)
VAN	bottle trap under the basin, 75 mm seal (visible)	Ø 40	2.87 / 4.65	2 / 3	+3.53	+3.43	yes, 0.04 spare	+3.92 (above +3.37)

Levels above the ground floor ±0.00; "bottom" = the pipe's outside bottom at the stack after the fall of 2 % (1:50) over the run; the joints: at the trap, at each bend and at the stack's branch fitting. The build-up: finish 0.08 (params.yaml), the depression 0.15 (structure.typ), tanking 3 mm + protection 20 mm, screed 50 mm, membrane 2 mm, tiles 14 mm: ASSUMED. The waterproofing: (1) a liquid tanking on the depressed slab turned 0.15 up the walls (the tray, the second line against leaks), (2) a bonded membrane under the tiles turned up the walls (the first line); in (b) the service wall stands on the tray inside the tanking, its cavity drained to the stack's branch through a weep (a leak in the cavity shows at the tray's tell-tale).

What the sections show

- **(a):** the pipe zone is 0.14 m; the WC's Ø 110 (an S-trap pan, its bend 0.07 under the tiles) fits with 0.02 m to spare over its 0.19 m run; the shower gully (0.09 body) and the vanity (0.07 bend) fit with 0.03 and 0.04 m to spare; **the tub fits only with a flat 0.08 trap on the straight diagonal under the shower** (0.02 m to spare): at the limit of the build-up, with a 0.10 trap or a run along the walls it does not fit at 2 %. Nothing hangs under the slab: **the kitchen ceiling carries only the lights** (the soffit +3.07 under the depressed zone, the gypsum at +2.9): also in (a). But every drain and joint is under 0.07 m of screed, membrane and tile.
- **(b):** no pipe below the tiles except the channel's body (0.10 deep in the screed, at the wall); the WC's outlet at +0.18, the channel's trap at +3.51 in the cavity, the tub's pipe from +3.62 to +3.51, the vanity's chase from +0.45 to +3.92: all above the tray, reached from the room side.
- **The kitchen ceiling under the bath:** in both options **nothing but the structure** (the ribbed slab, its depressed zone), the ceiling's lights and the HRV ducts of the kitchen itself (which run south of the hood zone). **No drain under the hood zone** (y ≥ 10.2): checked.
- **The tub by the window (proved):** its trap sits in the plinth at +3.62 … +3.72 the pipe falls 2 % over 3.79 m to +3.51 at the stack: about the finished floor's level, inside the service wall's cavity and 0.14 m above the tray; **no part of it is in the floor's screed or fill**. **Access:** the plinth's front panel (clip-on stone, 1.7 × 0.12) for the trap's cleaning and replacement; the pop-up waste from inside the tub. **Ventilation:** the branch is 3.79 m: within BS EN 12056-2's unvented 50 mm limit (4 m) but beyond the UPC's trap-arm limit, so an **air admittance valve in the plinth** (accessible, above the trap's weir) or a secondary vent up the service wall to the stack's vent: the engineer's code decides; with the AAV the trap cannot siphon.

A1 · Comparison (a) sunken slab · (b) service wall + plinth · (c) raised floor · the access of EVERY drain and joint · the recommendation

Criterion	(a) sunken slab, all buried	(b) service wall + tub plinth	(c) raised floor
Maintenance access	none without lifting tiles, screed and the second membrane (the tanking stays); blockages rodded from the fixtures and the stack's eye	the WC, the tub, the vanity's trap and all joints at the stack reachable without cutting anything; the channel's insert from the front; its body = the one local opening	the panels lift where dry; in the shower the sealing defeats it
Leak containment	the tanked tray holds the leak; it shows at the tray's tell-tale (late)	the same tray as the second line; the cavity and the plinth are dry voids with sensors (early)	the tray; the void is inspectable
Noise	pipes in the fill: quiet	the WC frame and the pipes in a cavity: lined with 25 mm mineral wool, the pipe acoustic; quiet	a hollow floor drums: needs a damping fill
Structure	the depressed zone exists (X10); no change	the same depressed zone; the service wall is light (steel frame + boards, ≈ 0.6 kN/m) on the slab; the bath loses 0.20 m in width (2.8 clear) and the closet 0.20 (0.9 clear: still a closet)	the same slab; the floor's pedestals
Relative cost	1.0	≈ 1.2 (the frame, the wall-drain channel, the plinth, the chase)	≈ 1.3 and worse in the shower
Function	as designed	a wall-hung WC and vanity (the owner's "high quality"); the tub on a plinth: a step of 0.12	a step in the floor

Recommendation: (b), with the honest limit

- **(b)**: the east service wall with the wall-hung WC, the wall-drain shower channel and the tub's plinth; the vanity's jointless chase. Every drain, trap and joint but one is reached from the room side or from the stack's cladding outside without cutting a finish; **the one exception is the shower channel's body** (its insert, strainer and seal come out from the front; the body itself, if it ever fails, needs a local opening ≈ 1.4 × 0.3 m of tiles and screed along the wall and the membrane re-made there, the tray below untouched). That is the smallest opening in the room and the best practical compromise: a channel body is a passive part with no moving element.
- **Keep the depressed tanked slab** in (b) too: it holds the channel's depth and is the second line against any leak; the cavity's weep and a tell-tale show a leak before it spreads. The kitchen ceiling under the bath carries nothing.
- **Also in (b):** the stack's four branch fittings and the rodding eye sit **outside, behind the cladding** (page 10): a blockage anywhere in the bath is cleared from the yard without entering the suite.

Every drain and joint (option b): buried or accessible, and how it is reached

Item	(1) blockage	(2) leak: shows first at · how reached	(3) broken pipe: replace	removed
WC pan + its seal	through the pan; the stack's eye outside	the pan's foot / the cavity sensor · the closet	unbolt the pan (2 bolts), the outlet bend through the frame's opening	nothing
WC cistern + valve	—	the flush plate's opening	through the flush plate (standard)	the plate
WC outlet bend → the wall → the stack	the stack's rodding eye	the cavity sensor; outside at the branch (cladding off)	from the closet after the pan, and from outside	the cladding panel
shower channel: grating, strainer, trap insert	lift the grating, pull the insert, rod the outlet	—	the insert is the wear part: swapped from the front	nothing
shower channel body (in the screed)	as above	the tray's tell-tale (the tanking holds it)	**the one local opening**: ≈ 1.4 × 0.3 m of tiles + screed, the membrane re-made	tiles + screed, locally
shower outlet Ø 50 in the wall cavity → the stack	the stack's eye; the channel's outlet	the cavity sensor; the front panel 0.6 × 1.0 above the channel (dry zone, tiled panel on a frame)	through the front panel	the panel
tub waste + pop-up	from inside the tub	the plinth void (sensor)	from the plinth	the plinth's front
tub trap + AAV (in the plinth)	the plinth's front panel	the plinth void	the plinth's front panel	the front panel
tub outlet Ø 50 in the plinth, the duct, the wall	the stack's eye	the duct's and the cavity's sensor; the duct's top is a lift-off stone shelf under the window	the duct's lid, the wall's front panel	the lid / a panel
vanity bottle trap	unscrew it (visible)	visible	visible	nothing
vanity Ø 40 in the north wall's chase	rod from the trap	the chase's foot at the wall's base above the floor; the cavity sensor	open the chase: a 0.10 m plaster strip (dry wall zone, no waterproofing)	plaster + tile strip
the four joints at the stack + the rodding eye	the eye	outside at the cladding's foot (a drip)	from outside, the panel off	the cladding panel

Sensors: a cable leak sensor in the service wall's cavity at the tray level, one in the plinth, one in the duct; the tray's tell-tale is a Ø 20 weep from the tanking's lowest corner through the wall into the stack's cladding zone (a drip there = a leak above). All wired to the house's automatic main shut-off valve.

A3 · The east stack: (a) the external stack on the east wall behind removable cladding vs (b) an internal accessible duct in the kitchen's north-east corner

Criterion	(a) external stack behind removable cladding (current)	(b) internal duct 0.35 × 0.35 in the kitchen's NE corner
Real access: where you stand, the panel, what you move	in the **service yard**, on the ground: a 0.6 × 2.4 m cladding panel (honey-stone slats on a frame, 4 clips) off the wall; the FF branch fittings at +3.4 … +3.9 reached from a 2-step platform or a ladder in the yard; the bins (0.8 m north) rolled aside; nothing inside the house touched	in the **kitchen**: a 0.6 × 2.1 m door in the corner; you stand in the kitchen's north-east corner, where the **service door (y 7.2–8.1) opens**: the duct's footprint (y 7.0–7.35) overlaps the door's jamb (reported): the door must move 0.4 m south (and the bins and the gas pipe with it); the tall storage / fridge beside it moved; the FF panel in the WC closet's corner (0.5 × 1.2) behind the WC: the WC's frame limits it
Noise	outside the wall: nothing heard inside; the pipe acoustic and insulated anyway	a soil stack in a kitchen wall: needs the acoustic pipe, a lined duct (mineral wool) and a heavy door; audible in the kitchen at night in the worst case
Odour and fire	the vent outside, up the wall to the roof; the branch joints outside: a leak or a smell stays outside; no fire separation needed	the duct is a flue through the kitchen roof slab: a fire collar at the slab, the duct fire-rated (60 min) on the kitchen side (the hood zone is 3 m south); a smell from a joint enters the kitchen
Weather and heat	the pipe behind the cladding: shaded, ventilated; UV-stable pipe; the east face gets the morning sun only (the yard's screen shades the foot)	none: inside
The east facade	the cladding panel is part of the east wall's service-yard face (the family's door, the bins, the screen): a 0.6 m stone-slat panel between the service door and the laundry's wall reads as a utility feature on the utility side; not on a show facade	nothing outside; but the service door moves
Structure	none: the stack hangs on the wall; the branches pass through the wall in sleeves (three Ø 110 / 50 holes in the 0.25 wall; no column at y 7.05: the east wall's columns stand at y 6.0 and 10.45)	a 0.35 × 0.35 opening through the ribbed kitchen roof slab at the corner, trimmed (the engineer); and the maid wing's solid slab is not involved
Relative cost	1.0 (the pipe, the cladding panel, the sleeves)	≈ 1.5 (the duct, the slab opening and trimmer, the fire collar, the door moved, the kitchen corner rebuilt)
What it helps	the vertical stack AND its four branch connections (all outside)	the vertical stack only: the branches still cross the service wall (page 9)

Recommendation: (a) the external stack behind removable cladding

- Round 1 leaned to the internal duct; the enlarged study **reverses that**: at the kitchen's north-east corner the duct collides with the service door and brings a soil stack, a slab opening and a fire-rated flue into the kitchen for a gain that is only the vertical pipe. The external stack puts the vertical AND every branch connection of the suite bath **outside the house**, reachable from the yard with a ladder, and keeps the kitchen's ceiling and corner clean.
- **Conditions for (a):** an acoustic, UV-stable Ø 110 pipe with a **rodding eye at the yard level and at +3.3** (under the branches); the four branch fittings at +3.4 … +3.9 behind one cladding panel 0.6 × 2.4 (clips, no tools); the cladding ventilated top and bottom; the vent carried 1.0 m above the roof (+8.2), 2.5 m above the bath window's head and 7.4 m from the HRV intake; the sleeves through the wall sealed both sides (fire and water); the stack's foot into the yard's foul chamber with a bend and a cleanout. The pipe's run on the wall is **inside the service yard's screen**: unseen from the garden and the lane.
- **The west cluster** keeps its stack inside the **shaft** (doors on both floors): the inside solution is right there because the shaft exists and is in a service room, not a kitchen.

What a duct would need, if the owner still wants it

- The service door moved to y 7.6–8.5 (the bins and the gas pipe with it), the kitchen's corner unit shortened by 0.4 m, a trimmed 0.35 × 0.35 slab opening (X10's zone), a 60-minute duct with a fire collar, a 0.6 × 2.1 access door in the kitchen and a 0.5 × 1.2 panel in the closet; and the branches still cross the service wall. Cost ≈ 1.5 × (a). Not recommended.

A4 · Every possible source of drain smell in the house and on the site, and how each is prevented (the owner's priority)

Source	Prevention (every item assumed, the plumbing engineer designs)
the foul stacks' vent terminals (the west shaft, the east stack)	open vents to the roof ≥ 1.0 m above the roof and ≥ 0.9 m above any opening within 3 m; ≥ 3 m from the HRV intake (7.4 / 12.5 m) and the malqaf; a bird cage; no AAV in place of the main vents
the WCs' and basins' traps	every fixture its own trap, 50 mm seal minimum (basins 75 mm bottle traps); no S-trap siphoning: the branches ventilated (the stacks' vents, a secondary vent or an AAV where a branch exceeds the unvented length)
the showers' and the floors' drains	the shower channel is the only floor drain in each bathroom (no separate floor gully to dry out); the channels' traps with a membrane valve; the rarely used floor drains (the utility room, the laundry, the shaft, the tower sump, the courtyard's step drain) get **anti-odour membrane traps** and a **trap primer** or a listed monthly fill
the tub's trap	a 50 mm seal in the plinth, used weekly; with the AAV it cannot siphon; the plinth panel vents to the room (no closed cavity)
the cleanouts and the rodding eyes (the stacks, the buried runs' bends)	sealed screw caps with gaskets, inside the shaft / behind the cladding; none in a habitable room
the inspection chambers (the yard, the driveway, the north strip)	sealed double-seal covers (recessed, gasketed) with the invert benched; the collector chamber by the gate the same; no open gully into the foul system outside
the septic tank (outside the wall near C1)	its vent ≥ 2 m high on the wall's outer side, away from the gate (6.8 m) and the north parking bay; its lid sealed; de-sludged on a schedule; no vent at the gate post
the kitchen's sink and the dishwasher	a deep trap (75 mm) with the dishwasher's hose above the seal; the grease kept out (a small grease trap under the sink is optional: the engineer); the hospitality kitchen the same
the laundry's washer standpipe	a trapped standpipe ≥ 0.6 m high, the hose not sealed into it (an air gap)
the maid bath and the guest WC (less used)	their traps still used weekly; the guest WC's floor is the shower-free type: no floor drain at all
the courtyard, the suite court (rain)	on the separate rain system, never connected to the foul one: no sewer gas can reach their drains; their gullies with leaf baskets
the condensate (AC, HRV), the tower sump, the drip trays	as the table on the round-2 corrections page: air gaps over tundishes or trapped air-breaks, never a direct joint to a drain or a vent
the pipes themselves	acoustic (triple-layer) soil pipes with push-fit sealed joints, pressure-tested before closing; the stacks never cut by a floor's finish: a sleeve and a fire collar at each slab

The rule behind the list: a smell needs an open path from the sewer to a room. The paths are a dry or siphoned trap, an unsealed cleanout or cover, a vent terminal near an opening or an intake, and a direct joint from a device to a drain. Each is closed above; the rarely used drains are the only ones that need care (a membrane valve and a primer or a listed fill), and the design keeps them few: no floor gully in any bathroom but the shower's channel.

The list for the owner (round 2)

1. **A1:** the suite bath as **(b)** (service wall, wall-hung WC and vanity, wall-drain channel, the tub on a plinth with an AAV): accepted? The one local opening (the channel body) accepted as the compromise?
2. **A3:** the east stack **external behind removable cladding** (not the internal duct): accepted?
3. **A2:** the rain system separate with two soakaways and surface overflows; the condensate by tundishes and trapped air-breaks: noted.
4. **A0:** the septic's position near C1 and the tanks east of peg R to be measured; the **septic's inlet level** decides whether a pump station is needed at the gate's collector chamber; the municipality's consent for the septic outside the wall.
5. The monthly maintenance list (the rarely used drains, the tower sump, the tundishes) accepted as a duty, or trap primers everywhere (more pipe, no duty)?

Study checks passed (16: round 1 9, round 2 7)

Round 2 checks: option a: every drain fits the sunken zone's pipe height (0.14 m from the protection screed +3.39 to the screed's underside +3.53); the deepest bottom at the stack 3.41 · option b: every pipe in the service wall, the duct or the plinth stays above the depressed slab (+3.37) and the tub's branch reaches the stack at about the floor level, inside the cavity (+3.51, the tray at +3.37) · no drain south of the hood zone's start (y 10.2) at the east wall: the tub's outlet at y 10 · the east stack's vent 2.5 m above the bath window's head and 7.4 m from the HRV intake · the tub's branch 3.79 m ≤ 4 m (BS EN 12056 unvented 50 mm limit) but > 1.07 m (the UPC trap-arm limit): an AAV in the plinth specified · the internal east duct (A3 option b) would overlap the kitchen's service door (y 7.2–8.1): reported · the rain points never drain to a foul collector; the condensate list gives every device a method. Every size is assumed, to be designed by the plumbing engineer; no estimate here (cost, length, depth) is proven; nothing in the room CSVs or the design files changes until the owner decides.